

Architecting the DSCR Sovereign OS: A Seven-Week Sprint to Resolve Eight Critical Gaps in TimesFM 2.5

This report details a systematic applied research and development plan designed to resolve eight critical technical gaps within a production-grade forecasting architecture built upon the TimesFM 2.5 foundation model. The objective is to harden this system, dubbed the 'DSCR Sovereign OS', into an institutional-grade, mathematically rigorous credit engine capable of withstanding regulatory scrutiny and extreme market volatility [77](#) [95](#). The architecture utilizes Low-Rank Adaptation (LoRA) to specialize the base model for real estate cashflow forecasting, with a focus on enabling accurate Monte Carlo stress testing for Debt Service Coverage Ratio (DSCR)-based credit decisions [20](#) [111](#). The proposed work program is structured as a seven-week sprint, progressing logically from foundational data topology and tokenization alignment to advanced mathematical stabilization, sophisticated modeling for spatial generalization and regime adaptation, and finally, production-grade inference optimization. Each phase directly addresses one or more of the identified gaps, culminating in a comprehensive validation framework and a final deliverable designed for regulatory defensibility. The solutions proposed are grounded in established principles of time-series analysis, transformer architecture, and modern MLOps practices, drawing from a wide range of academic and industrial sources to ensure both theoretical soundness and practical applicability [18](#) [70](#) [88](#).

Phase I: Data Topology and Tokenization Alignment

The initial phase of this research sprint is dedicated to establishing a robust and semantically correct data topology for feeding information into the TimesFM 2.5 model. This involves resolving three foundational issues: misalignment between the user-defined context window and the model's internal patching mechanism, improper separation of covariate channels, and the flawed handling of missing data points, colloquially known as "dark months." These gaps, if unaddressed, can silently corrupt the attention mechanisms and degrade the model's ability to learn meaningful temporal dependencies, irrespective of the underlying mathematics or model architecture. The primary goal of this phase is to

implement a strict data contract that adheres to the expectations of the TimesFM 2.0+ framework, ensuring that every piece of information provided to the model is correctly formatted, masked, and interpreted [18](#) [88](#). Success will be measured through a series of targeted ablation studies comparing model performance against a baseline configuration that does not incorporate these fixes. The core hypothesis driving this phase is that meticulous attention to data topology is a prerequisite for unlocking the full potential of the foundation model and preventing subtle but critical errors from propagating through the entire forecasting pipeline.

The first critical issue, **Gap 2: TimesFM Patching & Context Length Mismatch**, arises from the fundamental design of TimesFM, which employs a patch-based tokenizer similar to PatchTST [7](#) [35](#). The model operates on fixed-size patches of time-series data, and its attention mechanism is designed to operate within this patched representation. If the specified `context_len` for a forecasting task (e.g., 24 months of historical rent data) is smaller than the model's default `patch_len` (which is often 32 or 64), the input tensor will be heavily zero-padded at the beginning to meet the minimum patch requirement [1](#) [2](#). This excessive padding dilutes the effective signal-to-noise ratio of the context window. The attention mechanism, which is theoretically supposed to focus on salient temporal features, will instead be forced to process a large number of meaningless padded values, leading to a degradation of model accuracy and potentially unstable training dynamics [28](#). The proposed upgrade is to enforce a strict alignment between the `context_len` and the model's internal patch size. This can be achieved in two ways: either by adjusting the user-specified `context_len` to be a direct multiple of the patch size (e.g., using 32, 48, or 64 months of history) or by modifying the TimesFM configuration file in HuggingFace to use a smaller `patch_len` that better matches the available data granularity (e.g., setting `patch_len=12` to capture annual seasonality) [1](#). Critically, any implementation must include proper attention masking. The dataset loader must generate an `attention_mask` tensor where positions corresponding to valid data are marked as 1 and padded positions are marked as 0. This explicit mask informs the Transformer to ignore the padded regions entirely, preventing it from treating a padded month as a month with zero rent, which would catastrophically mislead the model [7](#). An empirical study found that masking out patches consisting entirely of padding values is an effective strategy for mitigating the negative impact of padding [7](#). To evaluate this change, an ablation experiment will be conducted by training four distinct LoRA variants of the model, each with a different `context_len`: 24, 32, 48, and 64 months. The performance of each variant will be benchmarked against the other on a held-out validation set. The primary success metric will be the Continuous Ranked Probability Score (CRPS), a proper scoring rule for evaluating probabilistic forecasts. A lower CRPS indicates a better-calibrated and more accurate predictive

distribution. Additionally, monitoring the entropy of the attention weights can provide a secondary insight; a well-aligned model should exhibit sharper, more focused attention patterns (lower entropy) compared to a diluted one [66](#).

The second foundational issue, **Gap 3: Covariate Routing (Past vs. Future Channels)**, stems from architectural updates in TimesFM 2.0 and later versions. These models have adopted a stricter, channel-aware design that mandates separate input tensors for past and future covariates [18](#) [88](#). The blueprint's approach of stacking all covariates into a single tuple violates this schema and will lead to silent corruption of the model's cross-attention mechanism. The model's encoder processes the `past_covariates` to build a memory of the historical context, while the decoder uses both the encoded past and the `future_covariates` to autoregressively generate predictions. Mixing these two types of information into one tensor disrupts this carefully orchestrated flow of information, likely introducing subtle lookahead bias where the model inadvertently uses information from the forecast horizon to predict the present [16](#). The upgrade requires a complete overhaul of the custom dataset and collator logic to conform to the dictionary-based input format expected by TimesFM. Specifically, the `__getitem__` method of the dataset class must return a dictionary containing explicitly separated tensors for `past_covariates` and `future_covariates`. The `past_covariates` tensor (`[batch, context_len, num_past_covs]`) should contain variables that are known only up to the end of the context window, such as trailing indicators of delinquency, historical vacancy rates, and STR occupancy levels [15](#). Conversely, the `future_covariates` tensor (`[batch, context_len + horizon_len, num_future_covs]`) can contain variables that are known for both the context and forecast horizons, including deterministic seasonal features (e.g., sin/cos of the month-of-year), known SOFR forward curves, scheduled tax hikes, and, most importantly, the cross-sectional macro anchors that will be discussed in the next phase [1](#). An ablation study will compare two model configurations: Model A, which uses the legacy mixed-tensor approach, and Model B, which implements the strict dictionary-based routing. The success of this change will be measured by comparing the out-of-sample Winkler Scores of the two models. The Winkler score evaluates both the accuracy of the central prediction and the width of the prediction interval; a significant improvement in calibration (i.e., a higher percentage of actual observed values falling within their predicted P10-P90 bands) for Model B would confirm that correct covariate routing leads to better-calibrated uncertainty estimates [16](#).

The third and final topological issue, **Gap 6: Handling "Dark" Months (Missingness)**, addresses the pervasive problem of incomplete data in real property management systems. It is common for rental properties to have months where no payment is recorded

due to renovation, vacancy, or administrative oversight. Treating these `null` or `0` values naively is a critical flaw. The sliding window approach will feed these zeros to the model, which will interpret them as a catastrophic market crash or a total loss of income, severely contaminating the training process and producing nonsensical forecasts. The recommended solution is a sophisticated two-pronged approach that separates the act of imputation from the communication of uncertainty. First, the raw rent time series should be preprocessed using a simple yet effective imputation strategy, such as forward-filling, to maintain continuity in the data stream fed to the model [61](#). Second, and more importantly, a binary Missingness Indicator Covariate must be introduced. This is a new feature vector added to the `past_covariates` that acts as a flag: it takes the value 1 if the corresponding rent observation was present and 0 if it was missing or imputed [10](#) [11](#). By providing this indicator, the model's attention mechanism is explicitly told when a data point is unreliable. The desired learned behavior is that during periods flagged as missing, the model automatically widens its P10-P90 confidence bands, accurately reflecting the heightened uncertainty associated with those months [9](#) [44](#). This approach is superior to simple mean or median imputation because it allows the model to learn a dynamic relationship between missingness events and forecast uncertainty. To test this, an experiment will be conducted comparing three imputation strategies for the rent values: (a) forward-fill (the proposed method), (b) filling with the MSA-level mean rent for that month, and (c) spline interpolation. The success metric will be the dynamically learned width of the P10-P90 confidence interval specifically during known periods of artificial missingness. The forward-fill method combined with the indicator covariate is expected to produce the most realistic and appropriately widened uncertainty bands, demonstrating that the model has successfully learned to reason about data quality [43](#) [48](#).

By systematically addressing these three foundational gaps in Phase I, the DSCR Sovereign OS will transition from a fragile pipeline susceptible to data-related artifacts to a robust system built on a solid data contract. The resulting data loading infrastructure will be aligned with the TimesFM model's internal mechanics, ensuring that the attention mechanism receives clean, correctly formatted, and semantically meaningful inputs. This sets the stage for the subsequent phases, which will introduce more complex mathematical and architectural innovations.

Gap	Problem Description	Proposed Solution	Success Metric
Gap 2: Patching Misalignment	Context window length is smaller than the model's patch size, causing excessive zero-padding that dilutes attention.	Align <code>context_len</code> to a multiple of the patch size or adjust <code>patch_len</code> . Implement an <code>attention_mask</code> to ignore padded values.	Minimize Continuous Ranked Probability Score (CRPS); Monitor attention weight entropy.
Gap 3: Covariate Routing	Past and future covariates are mixed in a single tensor, corrupting the cross-attention mechanism and risking lookahead bias.	Implement a strict dictionary format with separate <code>past_covariates</code> and <code>future_covariates</code> tensors as per the TimesFM schema.	Improve Winkler Score calibration (e.g., achieving ~90% of actuals within the P10-P90 band).
Gap 6: Missingness Handling	Missing data ("dark months") is treated as zero, confusing the model and leading to inaccurate forecasts.	Forward-fill rent values and introduce a binary <code>is_missing</code> indicator covariate to signal uncertainty to the model.	Dynamically widen the P10-P90 confidence interval during imputed periods.

Phase II: Mathematical Rigor and Probabilistic Calibration

With a stable data topology established, the second phase shifts focus to the core mathematical and statistical integrity of the forecasting model. This phase is dedicated to resolving three interconnected challenges: the insidious problem of log-return error compounding, the logical fallacy of quantile crossing in probabilistic outputs, and the overarching need for a mathematically sound approach to normalizing non-stationary financial time series. These issues are paramount for building a credit engine that can be trusted for high-stakes decision-making and stress testing. The solutions explored here move beyond heuristic fixes toward principled statistical frameworks that guarantee the mathematical consistency and reliability of the model's output. The central hypothesis is that leveraging the model's native capabilities and adopting state-of-the-art techniques for probabilistic forecasting are essential for transforming the DSCR Sovereign OS from a standard forecasting tool into a rigorously calibrated financial instrument. This phase will involve a comparative analysis of different mathematical formulations for the target variable and the output distribution, using metrics that directly measure the properties being addressed, such as tail risk and quantile monotonicity.

The first challenge, **Gap 1: The "Log-Return" Error Compounding Trap**, identifies a fundamental flaw in a seemingly intuitive forecasting strategy. The initial blueprint suggests forecasting the logarithmic returns of rent ($y_t = \log(\text{rent}_t / \text{rent}_{\{t-1\}})$) and then reconstructing the full rent path by exponentiating the cumulative sum of forecasts [80](#). While this transformation can stabilize variance over short horizons, it introduces a severe problem for multi-step forecasting. In quantitative

finance, variance is an additive property. Over a 12-month forecast horizon, the uncertainty (variance) of the cumulative sum of log-returns grows proportionally with the horizon. When this increasingly uncertain sum is exponentiated to convert it back to the original scale of raw rent, the additive growth of variance translates into multiplicative growth of tail risk [5](#). This causes the upper tail of the forecast distribution (e.g., the P90 quantile) to explode to unrealistic, astronomically high values, while the lower tail (P10) is crushed towards zero. This phenomenon renders the resulting Monte Carlo simulations useless for stress testing, as they no longer represent plausible scenarios but rather statistical artifacts of the compounding process [32](#). The proposed upgrade is to abandon manual log-transformations and instead rely on the TimesFM model's native Reversible Instance Normalization (RevIN) for stabilization [12](#). RevIN works by subtracting the instance-specific mean and dividing by the instance-specific standard deviation for each time series within the batch before it enters the model. After the model generates its output, an inverse normalization step restores the original scale [14](#). This process effectively makes the input data stationary for the purpose of learning, without forcing a global logarithmic scale. This is mathematically superior for preventing tail explosion over a 12-month horizon because it adapts the normalization parameters locally to the statistics of the context window, preserving the intrinsic variance structure of the data [12](#). However, this recommendation must be nuanced with recent findings. A comprehensive study revealed that standard RevIN contains several theoretical contradictions and can be fragile, particularly its reliance on non-robust statistics like the mean and standard deviation. On datasets with extreme outliers, naive RevIN can catastrophically fail, leading to massive increases in error [13](#). Therefore, a truly robust implementation might require a diagnostic-driven approach to choose the normalization strategy. Key diagnostics include assessing distribution skewness and the empirical k-factor (a measure of deviation from normality). Based on these diagnostics, one could select between sensitive methods like RevIN for relatively clean data or robust alternatives like R2-IN (a robust version of RevIN) for data with heavy tails or outliers [13](#). Despite these caveats, for typical real estate rent series, RevIN remains a powerful and pragmatic choice that avoids the pitfalls of manual transformations. To empirically validate this, an experiment will compare three target variable transformations: Variant A (log-returns, as baseline), Variant B (raw rent, relying solely on RevIN), and Variant C (MSA-detrended rent, calculated as $\text{Rent} / \text{MSA_Zillow_Rent_Index}$). The primary success metric will be the P90/P10 ratio at the 12-month horizon. The log-return method (Variant A) is expected to show an exploded ratio (e.g., >5.0), whereas Variants B and C should maintain a much more realistic and bounded variance (e.g., a ratio of 1.2 - 1.5), demonstrating successful control over tail risk [16](#).

The second challenge, **Gap 5: Quantile Crossing in the Continuous Head**, addresses a logical inconsistency that arises when using independent optimization for each quantile. Standard practice for generating probabilistic forecasts with deep learning models involves adding multiple output heads, each trained with the Pinball loss function to predict a specific quantile (e.g., P10, P50, P90) [60](#). The Pinball loss is strictly consistent, meaning it encourages the model to produce an output that minimizes the loss for its target quantile [60](#). However, because each head is trained independently, there is no guarantee that the outputs will be ordered correctly. It is possible, especially for highly volatile properties or in thin markets, for the model to predict a P90 quantile that is numerically lower than the P50 quantile (i.e., $P90 < P50$). Such a result is mathematically impossible for a true cumulative distribution function and breaks downstream applications that assume monotonicity, such as Monte Carlo simulations or the construction of coherent risk measures [47](#). The assistant proposes two solutions. The first is a simpler penalty-based approach: augment the overall loss function with a term that penalizes any crossing, such as $\text{Loss} + \lambda * \max(0, q_{\text{low}} - q_{\text{high}})$, where q_{low} and q_{high} are the predicted quantiles for the lower and higher probability levels, respectively [47](#). While straightforward to implement, this method can be brittle and difficult to tune the hyperparameter λ effectively. The second, preferred approach, is a more principled solution: replacing the discrete quantile heads with a single **Distributional Head** that predicts the parameters of a parametric probability distribution [16](#). The recommended distribution for this task is the Student's t-distribution, which is characterized by three parameters: location (μ), scale (σ), and degrees of freedom (ν). The model's final layer is simply modified to output these three parameters instead of multiple quantiles [16](#). Training is then performed by minimizing the Negative Log-Likelihood (NLL) of the observed data under the predicted Student's t-distribution. This approach is superior because it guarantees monotonic quantiles by construction; the quantiles are derived analytically from the fitted distribution parameters, ensuring they are always in the correct order [32](#). Furthermore, the Student's t-distribution is particularly well-suited for financial time series, as its heavy tails (controlled by the degrees of freedom parameter ν) can naturally accommodate the fat-tailed nature of real estate returns, which are prone to extreme events [80](#) [82](#). To evaluate this, an experiment will compare the "Crossing Rate"—the percentage of forecasts where $q_{\tau_1} > q_{\tau_2}$ for $\tau_1 < \tau_2$ —between the baseline quantile head model and the proposed distributional head model. The target for the distributional head approach is a crossing rate of exactly 0.0%, demonstrating perfect monotonicity in the output probabilities [84](#).

Together, these two solutions form a powerful combination for achieving mathematical rigor. By choosing a stable normalization scheme like RevIN and a guaranteed-monotonic

probabilistic output format like a Student-T distributional head, the DSCR Sovereign OS ensures that its forecasts are not only accurate but also logically consistent and statistically well-behaved. This level of mathematical fidelity is a cornerstone of an institutional-grade credit engine and is a critical step toward meeting the stringent requirements of regulators who demand robust model risk management frameworks ⁹⁵ ¹⁰¹. The final validation report for the sovereign OS will need to include detailed calibration backtests, akin to Hosmer-Lemeshow tests for classification models, to demonstrate that the predicted quantiles match their empirical coverage probabilities across the entire forecast distribution ⁷⁷.

Phase III: Spatial Generalization and Cross-Sectional Anchoring

Phase III addresses the critical need for the DSCR Sovereign OS to generalize beyond the idiosyncratic history of an individual property and understand the broader economic forces that govern real estate markets. This phase focuses on resolving **Gap 4: Spatial Blindspots & Cross-Sectional Anchors**, which describes a fundamental limitation of treating each property as a completely isolated forecasting unit. Without external guidance, the model learns local patterns from its limited context window but fails to grasp the concept of "contagion" or shared macroeconomic shocks that affect an entire Metropolitan Statistical Area (MSA) ⁴ ²³. This forces the model to relearn basic economic gravity for every single property, severely hampering its ability to make reliable forecasts for new properties in unfamiliar geographies or during novel market-wide events. The proposed solution is to inject **Cross-Sectional Macro Anchors** into the model's input, providing it with a crucial link to the wider economy. This technique effectively bridges the gap between univariate models that treat each entity independently and true multivariate models that explicitly capture cross-sectional dependencies ¹ ⁹⁶. The core hypothesis is that by giving the attention mechanism a strong, consistent signal from the macro-environment, the model can learn transferable features and achieve vastly improved out-of-sample geographic generalization. The success of this intervention will be validated through a rigorous Geographic Hold-Out Test, measuring the reduction in error on geographically unseen MSAs.

The architectural innovation lies in how external information is integrated. Instead of trying to build a complex, fully coupled multivariate model that feeds all properties' data simultaneously—a computationally expensive and architecturally challenging endeavor—

the proposed solution is more elegant and efficient ¹. It involves injecting key macroeconomic indicators for an MSA as a *future covariate* for every single property located within that same MSA. The most suitable candidates for these anchors are publicly available, high-frequency datasets that track local market conditions. The primary anchor is the MSA-level Zillow Rent Index (ZRI), which provides a normalized, repeat-sales index of rent changes across the MSA ⁴. Secondary anchors could include the MSA-level vacancy rate or other relevant indices of economic health ²³. These macroeconomic time series are concatenated with the property-specific past and future covariates and fed into the `future_covariates` channel of the TimesFM model. This gives the attention mechanism a clear "macro anchor" to latch onto. During training, the model learns a powerful correlation: *"when the MSA-level ZRI shows a sharp decline, this specific property's rent trajectory is highly likely to follow suit, even if its own recent history doesn't yet reflect that trend."* This imbues the model with a form of domain knowledge about real estate market dynamics, allowing it to generalize far more effectively. The effect is akin to teaching the model the "grammar" of how local markets behave in response to national or regional economic cycles ¹.

To validate the effectiveness of this approach, a **Geographic Hold-Out Test** will be conducted. This is a more stringent evaluation than a simple time-series split. The procedure is as follows: 1. **Data Partitioning:** All properties are grouped by their MSA identifier. A random subset of MSAs (e.g., 10%) is reserved as a hold-out set. The model is trained exclusively on data from the remaining 90% of MSAs. 2. **Training:** The model is trained using the full suite of covariates, including the newly injected MSA-level ZRI and vacancy rates as future anchors. 3. **Validation:** The trained model is evaluated on the hold-out set of MSAs. The forecasts for properties in these unseen MSAs are compared against their actual rent trajectories. 4. **Comparison:** The performance of the model with macro anchors will be compared against a control model that was trained identically but without the MSA-level covariates.

The primary success metric for this experiment will be the Mean Absolute Percentage Error (MAPE) on the hold-out MSAs. The hypothesis is that the model with macro anchors will exhibit a significantly lower MAPE compared to the model without them. This reduction in error demonstrates that the anchors have successfully transferred knowledge from familiar markets to unfamiliar ones, improving the model's spatial generalization capabilities. This is a critical capability for a sovereign AI system intended to operate across a diverse portfolio of properties in numerous geographies. By anchoring the local, micro-level predictions to a robust, macro-level signal, the DSCR Sovereign OS becomes less of a black box memorizing individual property histories and more of a transparent, reasoning engine grounded in the fundamental economic realities of the

housing market. This approach aligns with the principle of incorporating explanatory variables to leverage additional information, a practice shown to improve forecasting performance in various domains, including retail demand forecasting where calendar and price data are used to great effect [15](#) [16](#). The integration of these cross-sectional anchors transforms the model from a collection of parallel univariate predictors into a cohesive system that understands its place within a larger, interconnected market ecosystem.

Phase IV: Regime-Adaptive Forecasting with Specialized Adapters

Phase IV confronts the most sophisticated challenge among the eight gaps: **Gap 7: Regime-Blended LoRA (The "Sovereign" Edge)**. Financial markets, and real estate markets in particular, are not static; they operate in distinct economic regimes characterized by different interest rate environments, inflation dynamics, and liquidity conditions [5](#). A monolithic LoRA adapter, trained on a long, continuous period that spans multiple regimes (e.g., 2015–2025, covering both a low-rate era and a high-rate/inflationary era), will inevitably struggle. The optimization process will attempt to find a single set of adapter weights that represents an average of the dynamics from both regimes. This results in a "mushy middle" that is highly accurate for neither environment. The model will fail to capture the unique behavioral patterns of either the low-rate or high-rate periods, leading to poor performance when the current economic conditions shift. The proposed solution is a paradigm shift in the fine-tuning strategy: instead of training a single generic adapter, the team will train **Multiple Specialized Adapters** and develop a mechanism to dynamically blend them at inference time based on the prevailing macroeconomic regime [51](#). This creates a regime-aware model that can adapt its forecasting behavior to the current economic climate without requiring continuous retraining, embodying the "sovereign" aspect of the DSCR Sovereign OS by making it resilient to structural economic changes.

The implementation of this strategy involves a multi-step process. First, the training data will be partitioned chronologically to isolate the distinct regimes. Two specialized LoRA adapters will be trained:

- **Adapter A (LoRA-Zero):** This adapter will be trained exclusively on data from the Low-Rate Regime, defined as the period from 2015 to early 2022. This data captures a market environment of historically low interest rates, steady

appreciation, and high liquidity, which influences tenant demand, refinancing behavior, and property values in specific ways ⁵.

- **Adapter B (LoRA-One):** This adapter will be trained exclusively on data from the High-Rate/Inflation Regime, covering the period from 2022 to the present. This data reflects a market characterized by rising interest rates, increased borrowing costs, potential cooling of demand, and greater economic uncertainty ⁵.

Both adapters will be initialized with identical hyperparameters (e.g., rank $r=16$, alpha $\alpha=32$) and will be applied to the same layers of the frozen TimesFM base model (e.g., q_proj and v_proj in the last few transformer layers) to ensure they are learning comparable representations of the underlying dynamics ¹ ². The rationale is that by training each adapter on a more homogeneous dataset, it can learn the specific patterns and relationships characteristic of its designated regime without being confused by contradictory signals from another.

The second, and more innovative, part of the solution is the implementation of a **Dynamic Weight Blending** mechanism at inference time. This is the core of the "regime-aware" capability. When a new forecast is required, the system will not just apply a single set of adapter weights. Instead, it will compute a final set of weights by interpolating between the base model weights, the weights of Adapter A, and the weights of Adapter B. The formula for this is expressed as:

$$W_{final} = W_{base} + \gamma \cdot W_{Zero} + (1 - \gamma) \cdot W_{One}$$

Here, W_{base} represents the frozen weights of the pre-trained TimesFM model, while W_{Zero} and W_{One} are the trained weights of the two specialized adapters. The blending factor, γ , is a scalar between 0 and 1 that determines the influence of each adapter. The key question is how to determine γ in real-time. The proposal is to define γ using a simple heuristic gating function based on a readily available, high-frequency macroeconomic indicator. For example, a rule could be implemented: if the current 10-Year Treasury yield is below 3.5%, γ is set to 1.0 (favoring the low-rate regime adapter); if the SOFR rate is above 4.0%, γ is set to 0.0 (favoring the high-rate regime adapter); and for values in between, a linear interpolation is used. This approach is inspired by online adaptation techniques but offers a simpler, more interpretable alternative ⁸. The ADAPT-Z method, for instance, proposes online adjustment of latent feature representations, which is a more complex adaptive system ⁸. For a sovereign AI, the transparency and determinism of the proposed blending mechanism may be more valuable for regulatory review than a continuously adapting system whose internal logic is harder to audit.

The success of this advanced adapter topology will be rigorously tested. The primary experiment will compare the performance of the blended model against a strong baseline: a single, monolithic LoRA adapter trained on the entire 2015–2025 dataset. Both models will be evaluated on an out-of-sample test set comprising vintages from 2023–2024, a period that straddles the regime shift and is therefore challenging for the monolithic model. The primary success metrics will be the Continuous Ranked Probability Score (CRPS) and the Pinball Loss, evaluated separately for the P10, P50, and P90 quantiles. The hypothesis is that the blended model will achieve a lower CRPS and Pinball Loss, indicating a more accurate and better-calibrated forecast. This phase represents a significant architectural evolution for the DSCR Sovereign OS, moving it from a static model to a dynamic system capable of navigating the inherent non-stationarity of financial time series. By creating and intelligently combining specialized experts, the system gains a crucial form of resilience, a hallmark of a truly sovereign-grade AI.

Phase V: Inference Optimization for Real-Time Underwriting

The final phase of the sprint is dedicated to production readiness, focusing squarely on **Gap 8: Excessive Inference Latency**. While the preceding phases have been concerned with the accuracy and mathematical correctness of the model, this phase addresses its operational viability. Running a 500-million-parameter transformer model through a native Python implementation via the `HuggingFace Trainer` is fundamentally too slow for the intended use case of a real-time underwriting API that must process thousands of loan applications per minute [6](#). The current estimated latency of ~200ms per property is prohibitive, as it would make interactive, browser-side Monte Carlo simulations with 10,000 paths impossible while a borrower is on the phone [1](#). The goal of this phase is to reduce this latency by a factor of ten or more, targeting a sub-20ms inference time per property, thereby unlocking the system's full potential for dynamic, real-time decision support. The solution is a well-established, modern MLOps workflow involving model serialization, graph optimization, and hardware-specific compilation.

The first step in this optimization pipeline is **ONNX Export and Graph Surgery**. ONNX (Open Neural Network Exchange) is an open-standard format for representing machine learning models. Exporting the finalized, merged TimesFM model (with LoRA weights incorporated into the base model) to the ONNX format is the critical first step toward

cross-platform deployment and optimization [70](#). This is accomplished using the `torch.onnx.export` function, which traces the model's computation graph and saves it in a portable format. During this export, it is important to specify dynamic axes for dimensions that vary at inference time, such as batch size and sequence length, to maximize flexibility [70](#). Once the model exists in the ONNX format, a process of "graph surgery" can be performed. This involves inspecting the ONNX graph and removing any nodes that were required for training but are unnecessary for inference, such as certain dropout operations or backward-pass computations related to normalization layers [70](#). The goal of this surgical cleanup is to create a leaner, more efficient computational graph that is tailored specifically for the forward pass.

The second and most impactful step is **TensorRT Compilation**. NVIDIA TensorRT is a high-performance deep learning inference optimizer and runtime library. It takes the serialized ONNX graph and performs a suite of aggressive optimizations tailored to run on NVIDIA GPUs. These optimizations include: layer and tensor fusion (combining multiple operations into a single kernel launch to reduce overhead), kernel auto-tuning (selecting the fastest available implementation for each operation), and precision calibration. The most significant benefit comes from quantization. TensorRT supports running models in reduced precisions like FP16 (half-precision floating point) and INT8 (8-bit integer). Converting the model to FP16 can dramatically reduce memory bandwidth requirements and increase computational throughput on modern GPUs, typically yielding a latency reduction of over 80% with minimal to no loss in forecast accuracy [69](#) [70](#). While INT8 quantization offers even greater potential speedups, its performance can be hardware-dependent, and in some cases, an INT8 engine compiled with TensorRT can be slower than its FP16 counterpart, depending on the specific GPU architecture and the model's sensitivity to quantization error [71](#). Given the target of sub-20ms latency, starting with FP16 is the most prudent strategy. The final output of this step is a highly optimized `.engine` file, which is a serialized, platform-specific representation of the optimized model ready for execution.

The final stage of this phase is **Benchmarking and Validation**. Before deploying the optimized engine, it must be rigorously tested against the original PyTorch implementation to ensure accuracy parity. This is a non-negotiable step for a system intended for financial risk assessment. A load test will be conducted using a tool like Locust or the Triton Inference Server to simulate a high-volume production workload. The primary success metrics for this phase are:

- **Latency:** The average end-to-end latency for a single-property inference (batch size of 1) must be less than 20 milliseconds.

- **Throughput:** The system must be able to sustain a throughput of greater than 500 properties per second under load.
- **Accuracy Parity:** To ensure the optimization did not introduce numerical drift, the maximum absolute difference between the outputs of the original FP16 PyTorch model and the TensorRT FP16 engine must be below a very strict threshold, such as 1×10^{-3} , for a representative sample of inputs. This cryptographic or statistical parity check proves that the production inference engine is functionally equivalent to the research model [70](#).

Successfully completing this phase completes the journey from a research prototype to a production-grade system. The resulting sovereign OS will not only be mathematically sound and adaptable but also fast enough to power a truly interactive and responsive underwriting experience, fulfilling the core promise of the project. The final deliverable will include this optimized TensorRT engine, forming the operational backbone of the DSCR Sovereign OS.

Synthesis and Regulatory Readiness

This applied R&D sprint, structured over seven weeks, systematically deconstructs and resolves eight critical technical gaps that stand between a promising theoretical architecture and a production-grade, sovereign-level AI for real estate finance. The progression from foundational data topology to advanced mathematical and architectural innovations, culminating in production optimization, forms a coherent and logical pathway to building the DSCR Sovereign OS. The solutions proposed are not merely incremental tweaks but represent a holistic hardening of the system against the specific failure modes encountered when applying a powerful, general-purpose foundation model to the high-stakes domain of financial forecasting. By addressing the problems of log-return error compounding, patching misalignment, improper covariate routing, spatial blindspots, quantile crossing, data missingness, regime-agnostic training, and inference latency, the resulting system transitions from a "clever forecasting pipeline" to an "institutional-grade, mathematically rigorous credit engine" [77](#) [95](#).

The synthesis of these efforts yields a model that is fundamentally more robust and trustworthy. The adoption of TimesFM's native Reversible Instance Normalization (RevIN) provides a mathematically sound method for stabilizing non-stationary data, avoiding the tail-risk explosion inherent in manual log-transforms [12](#) [14](#). The implementation of a strict data contract, with explicit attention masking and correct

routing of past and future covariates, ensures the model's attention mechanism receives clean, semantically meaningful information [7](#) [18](#). The introduction of a Missingness Indicator Covariate allows the model to probabilistically reason about data quality, dynamically adjusting uncertainty estimates where data is sparse or absent [10](#) [11](#). Furthermore, the injection of cross-sectional macro anchors, such as the MSA-level Zillow Rent Index, imbues the model with crucial domain knowledge, enabling it to generalize across geography and navigate market-wide shocks with greater fidelity [4](#). At the heart of its probabilistic capabilities, the switch to a Student's t-distributional head guarantees monotonic quantiles by construction, eliminating logical inconsistencies and ensuring mathematical coherence for all downstream applications, including complex Monte Carlo simulations [16](#) [80](#).

Perhaps the most strategic enhancement is the development of a regime-adaptive forecasting engine through the training and dynamic blending of specialized LoRA adapters [51](#). This architecture directly confronts the non-stationarity of financial markets, allowing the DSCR Sovereign OS to pivot its forecasting behavior in response to changing macroeconomic environments, such as the shift from a low-rate to a high-rate world [5](#). Finally, the rigorous application of the ONNX and TensorRT workflow delivers a production-grade inference engine capable of sub-20ms latency, making real-time, interactive risk assessment a tangible reality [70](#). This end-to-end process, from data cleaning to low-latency deployment, is meticulously documented and validated throughout the sprint.

To ensure this engineered system is not only performant but also defensible to regulators, a formal governance and evidence vault integration process is paramount. Every experiment, from data lineage to hyperparameter tuning, must be logged with cryptographic integrity [77](#). The final deliverable is not merely the optimized model but a comprehensive **DSCR Sovereign OS Model Card** [101](#). This document must serve as the definitive source of truth, containing: 1. The complete, validated data transformation and preprocessing pipeline. 2. The final, merged, and TensorRT-compiled `.engine` file for inference. 3. The Python inference wrapper code, which includes the dynamic regime-blending logic (γ calculation). 4. A comprehensive validation report that provides irrefutable evidence of the system's properties, including:

- **Mathematical Stability:** Proof of no quantile crossing and a bounded P90/P10 ratio, demonstrating control over tail risk.
- **Calibration Backtests:** Automated generation of Hosmer-Lemeshow-style plots for the quantile outputs, proving that the model's uncertainty estimates are well-

calibrated (e.g., the P10 quantile is exceeded by approximately 10% of actual observations).

- **TensorRT Parity Reports:** Cryptographic hashes or statistical tests proving that the optimized inference engine produces bit-for-bit or numerically equivalent results to the original research model, confirming that no accuracy was lost in the optimization process [70](#).

By adhering to this rigorous process, the DSCR Sovereign OS emerges not just as a technological achievement but as a responsible AI system. It embodies the principles of a risk-based approach to model risk management, where the level of rigor applied is commensurate with the materiality of the model's use [95](#). The resulting sovereign OS is a system that is not only capable of making accurate forecasts but is also transparent, auditable, and demonstrably robust, making it a worthy candidate for deployment in a regulated financial environment.

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